Issue date: 10/8/2003
Revision date: 4/6/2023

Safety Data Sheet

1. Chemical product and company identification

Product name : Formamide

Company information

Name of manufacturer	:	KANTO CHEMICAL CO., INC.
Address	:	2-1, Nihonbashi, Muromachi 2-Chome, Chuo-Ku, Tokyo, 103-0022, JP
Name of section	:	Business Administration Department, Reagent Division
Telephone number	:	+81-3-6214-1090
Facsimile number	:	+81-3-3241-1047
Mail address	:	BC32@kanto.co.jp
Reference No	:	16062
Product numbers applied by the SDS	:	16062, 10055-08

2. Hazards identification

GHS classification

Physical hazards	Corrosive to metals	Category 1
Health hazards	Carcinogenicity	Category 2
	Reproductive toxicity	Category 1B
	Specific target organ toxicity (single exposure)	Category 3 (narcosis)
	Specific target organ toxicity (repeated exposure)	Category 2 (reproductive organs (male))

Hazard pictograms



Signal word : Danger

Hazard statements : May be corrosive to metals
May cause drowsiness or dizziness
Suspected of causing cancer
May damage fertility or the unborn child
May cause damage to organs (reproductive organs (male)) through prolonged or repeated exposure

Precautionary statements

Prevention : Do not handle until all safety precautions have been read and understood.
Keep only in original container.
Do not breathe mist/vapors.
Use only outdoors or in a well-ventilated area.
Wear protective gloves/protective clothing/eye protection/face protection.

Response : IF INHALED: Remove person to fresh air and keep comfortable for breathing.
IF exposed or concerned: Get medical advice/attention.
Call a POISON CENTER or doctor if you feel unwell.

		Get medical advice/attention if you feel unwell. Absorb spillage to prevent material-damage.
Storage	:	Store in a well-ventilated place. Keep container tightly closed. Store locked up.
Disposal	:	Dispose of contents/container to hazardous or special waste collection point, in accordance with local, regional, national and/or international regulation.

3. Composition/information on ingredients

Distinction of substance or mixture : Substance

Chemical name	Concentration (%)	Formula	TSCA	EC-No.	CAS RN
Formamide	≥ 98	CH3NO	Listed	200-842-0	75-12-7

4. First aid measures

First aid measures

First-aid measures after inhalation : Remove the victim to fresh air, and make him blow his nose and gargle.

First-aid measures after skin contact : Wash the affected areas under running water.

First-aid measures after eye contact : Wash the affected areas under running water for at least 15 minutes. If necessary, get medical treatment.

First-aid measures after ingestion : The chemical is volatile. Do not induce vomiting because it increases the risk of aspiration into the lungs. Get medical attention immediately. If necessary, rinse mouth with water.

Personal Protection in First Aid and Measures : Rescuers should wear proper protective equipment like rubber gloves, goggles.

Most Important Symptoms/Effects

Symptoms/effects : Inhalation causes lethargy, headache, nausea, unconsciousness.

5. Fire fighting measures

Suitable extinguishing media : Water, dry chemical powder, carbon dioxide, dry sand, alcohol resistant foam

Unsuitable extinguishing media : Foam extinguisher

Firefighting instructions : Move containers from fire area if it can be done without risk, if not possible, apply water from a safe distance to cool and protect surrounding area.

Dry chemical powder, carbon dioxide or dry sand should be used for small fires. Alcohol-resistant foam extinguisher is effective for a large scale fire.

Personal protection (Emergency response) : Wear breathing apparatus.

6. Accidental release measures

Personal Precautions, Protective Equipment and Emergency Procedures

General measures	:	Wear proper protective equipment and avoid contact with skin and inhalation of vapor. Conduct operations from upwind and evacuate people downwind. Remove all sources of ignition. Keep away personnel except for authorized ones from spillage area by stretching ropes.

Environmental precautions

Environmental precautions	:	Attention should be given to avoid discharge of spilled product into rivers and resulting environmental damage. When diluting spill with large amounts of water, discharge of untreated wastewater into the environment must be avoided.

Methods and Equipment for Containment and Cleaning up

For containment	:	Absorb spill with inert material (e.g, diatomaceous earth, sand) and flush spillage area with copious amounts of water.
Prevention Measures for Secondary Accidents	:	Remove nearby sources of ignition and prepare extinguishing media.

7. Handling and storage

Handling

Technical measures	:	Wear proper protective equipment to avoid contact with skin or inhalation of vapor. Pay attention to fire. Ventilate well at working places.
Precautions for safe handling	:	Use with an enclosed system or a local exhaust ventilation. Use in well-ventilated areas. Do not allow contact with oxidizing substances.

Storage

Storage conditions	:	Store the bottle tightly closed in a cool, dark place because the substance is hygroscopic.
Material used in packaging/containers	:	Glass, fluorine resin, stainless steel. Do not use polyvinyl chloride resin, polystyrene.

8. Exposure controls / Personal protection equipment

ACGIH TWA	10 ppm
Remark (ACGIH)	Skin

Appropriate engineering controls	:	Use with an enclosed system or a local exhaust ventilation.

Protective equipment

Respiratory protection	:	Chemical cartridge respirator with an organic vapor cartage or airline respirator
Hand protection	:	Impervious protective gloves
Eye protection	:	Safety goggles
Skin and body protection	:	Protective clothing, protective boots

9. Physical and chemical properties

Physical state	:	Liquid
Color	:	Colorless.
Odor	:	Ammonia like

pH	:	Weak alkalinity
Melting point	:	2.55 ° C
Freezing point	:	No data available
Boiling point	:	210.5 ° C (Partially disassembled)
Flash point	:	175 ° C (C.C.)
Auto-ignition temperature	:	> 500 ° C
Decomposition temperature	:	No data available
Flammability	:	Flammable
Vapor pressure	:	0.004 hPa (20℃)
Relative density	:	No data available
Density	:	1.133 - 1.138 g/cm³ (20℃)
Relative gas density	:	1.6
Solubility	:	Water: Miscible. Organic solvents: Miscible with ethanol, acetone.
Partition coefficient n-octanol/water (Log Pow)	:	-1.51
Explosive limits (vol %)	:	2.7 - 19 vol %
Viscosity, kinematic:	:	3.31 mm²/s (20℃)
Particle characteristics	:	No data available

10. Stability and reactivity

Reactivity	:	When hydrolyzed, it becomes ammonium formate, but when it is dehydrated by heating, it becomes formamide again. Corrodes copper, brass and mild steel.
Chemical stability	:	Stable under normal conditions. Hygroscopic.
Possibility of hazardous reactions	:	N, N-dichloromethane produced by reacting with hypochlorous acid is explosive. Hydrogen cyanide is produced by a powerful dehydrating agent such as phosphorus pentoxide.
Conditions to avoid	:	Light, heat, moisture.
Incompatible materials	:	Oxidizing substances.
Hazardous decomposition products	:	Carbon monoxide, nitrogen oxides, hydrogen cyanide.

11. Toxicological information

Acute toxicity (oral)	:	No classification rat LD50=3200mg/kg
Acute toxicity (dermal)	:	No classification rabbit LD50=6000mg/kg
Acute toxicity (inhalation)	:	No classification (gas) Classification not possible (vapor) No classification (mist) rat LC50>21mg/L/4h
Skin corrosion/irritation	:	No classification The substance has been reported to be slightly irritating to skin and eyes.

Serious eye damage/irritation	:	No classification In a rabbit eye irritation test (corresponding to OECD TG405), the overall average scores for 24/48/72 hours were 1.91 for conjunctival redness, 0.44 for edema, and 0.17 for corneal opacification. Based on the above, it has been reported that this substance is slightly irritating to the eyes of rabbits.
Respiratory sensitization	:	Classification not possible
Skin sensitization	:	Classification not possible
Germ cell mutagenicity	:	Classification not possible As for in vivo, it was negative in a dominant lethal test with mice, negative in a micronucleus test with mouse peripheral blood erythrocytes, and positive in a mouse bone marrow micronucleus test by intraperitoneal administration. As for in vitro, it was negative in bacterial reverse mutation tests. Micronucleus test findings are conflicting, with multiple oral doses being negative and single intraperitoneal doses being positive. The micronucleus inducibility was unclear due to the limitations of each test, and it was classified as "classification not possible" due to lack of data.
Carcinogenicity	:	Suspected of causing cancer ACGIH classifies it as the group A3 (confirmed animal carcinogen with unknown relevance to humans).
Reproductive toxicity	:	May damage fertility or the unborn child There is a report that in a continuous breeding study with mice by the oral route, fertility effects were observed at a dose where parental toxicity. There is a report that in a teratogenicity test with mice by the oral route, skeletal malformations were observed in fetuses at a dose where no maternal toxicity was observed. From the above, it was classified into category 1B.
STOT-single exposure	:	May cause drowsiness or dizziness In a single inhalation exposure test using rats, symptoms of "lethargy, hunchback posture, clear or red eye discharge, red nasal discharge, partially closed eyes, diarrhea, and brown staining of the lower abdomen" were observed at doses of 14-21 mg/L. It is reported that the symptoms almost disappeared on the 8th day after exposure. Based on the above, it was classified into category 3 (narcosis).
STOT-repeated exposure	:	May cause damage to organs (reproductive organs (male)) through prolonged or repeated exposure In a 2-year combined chronic toxicity/carcinogenicity study by oral gavage in rats and mice, hyperplasia of bone marrow in rats, calcification of testicular arteries and sheath of testis and spleen in mice at 80 mg/kg/day. It has been reported that hematopoietic cell proliferation was observed. There is also a report that blood effects were observed at 300 mg/kg/day in two 90-day repeated dermal administration studies using rats. Furthermore, in a 2-week repeated inhalation exposure test using rats, it was reported that platelet count decreased at 500ppm or more, and kidney effects and testicular degeneration were observed at 1500ppm. Based on the above, the substance was classified into category 2 (reproductive organs (male)) because effects on testis were observed within the dose range of category 2.
Aspiration hazard	:	Classification not possible

12. Ecological information

Ecotoxicity

Aquatic acute	:	No classification Pseudokirchneriella subcapitata ErC50>1000mg/L/72h
Aquatic chronic	:	No classification Pseudokirchneriella subcapitata NOEC>10mg/L/72h

Persistence and degradability

Readily biodegradable
BOD : 99%

Bioaccumulative potential

Low bioconcentration
log Pow : -1.51

Mobility in soil

High mobility
Koc : 3.6

Hazardous to the ozone layer

Ozone	:	Classification not possible

13. Disposal considerations

Ecology - waste materials	:	Burn in a chemical incinerator equipped with an afterburner and a scrubber. Or entrust approved waste disposal companies with the disposal.
Contaminated container and packaging	:	In case of disposal of empty bottles, dispose bottles after removing the content thoroughly.

14. Transport information

International Regulations

Transport by sea(IMDG)

UN-No. (IMDG)	:	1760
Proper Shipping Name (IMDG)	:	CORROSIVE LIQUID, N.O.S.
Packing group (IMDG)	:	III
Transport hazard class(es) (IMDG)	:	8

Air transport(IATA)

UN-No. (IATA)	:	1760
Proper Shipping Name (IATA)	:	Corrosive liquid, n.o.s.
Packing group (IATA)	:	III
Transport hazard class(es) (IATA)	:	8

Marine pollutant	:	Not applicable

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Pollutant category	:	Y
MFAG-No	:	154

15. Regulatory information

Regulatory information with regard to this substance in your country or region should be examined by your own responsibility.

16. Other information

Data sources : Solvents Handbook, T, Asahara el, Kodansha Scientific Ltd. (1976) .
Dictionary of Organic Compounds, The society of Synthetic Organic Chemistry, Kodansha Ltd. (1985) .
Handbook of 17322 Chemical Products, The Chemical Daily Co. (2022) .
NITE Chemical Risk Information Platform (NITE-CHRIP), National Institute of Technology and Evaluation.

The information contained herein is based on several references and the present state of our knowledge. However the SDS does not always cover all information about the product, handle the product carefully.The information is intended to ordinary usage, in case of particular handlings, conduct appropriate safety measurements.The information herein is only provision of information, and it does not represent a guarantee the properties of the product.The Safety Data Sheet(SDS) is prepared based on JIS Z7253.